

Web Development Intern Assignment — Round 1

Welcome

Thanks for applying to TechVruk. Because of the volume of applications, our hiring process is structured as:

1. **Round 1 (this document)** — 7-day take-home build. AI-assisted review at first pass, human review for shortlisted submissions.
2. **Round 2** — A shorter, focused build (~2 days). Released within 3 working days of the Round 1 deadline, only to candidates who clear Round 1.
3. **Interviews** — 1–2 conversations with the team, including a code walkthrough.

Because we use AI to review submissions at the first stage, **the format of your deliverable matters as much as the work itself**. Follow the file naming and structure exactly. Submissions that don't follow the format are filtered out before any human sees them.

This is not a coding assignment. It's an entrepreneurial assignment that requires shipping working code. We're hiring people who can ship a product, not people who can solve a Leetcode problem.

Timeline

- **Released:** Date on which shared
- **Deadline:** + 7 days
- **Late submissions are not reviewed.** No exceptions.

What we're evaluating

What	How we measure it
Discipline	Did you start early? Are commits and devlog entries spread across the week?
Hard work	Depth, polish, and how much you actually shipped relative to scope
Consistency	Daily progress over multiple distinct calendar days, not weekend cramming
Programming skills	Idiomatic, readable, debuggable code
Thinking models	Quality of trade-offs, decisions, architecture writeups, and reflection
Engineering skills	Git hygiene, tests, CI, docs, accessibility, deployed and working
Entrepreneurial thinking	Do you understand the user, the economics, and how to ship something people will actually use?

The Project — AI Spend Audit

The opportunity (read this carefully)

here's the problem: **most startups don't know they're overspending on AI tools in the first place.** They look at their monthly bill, sigh, and pay it. They have no benchmark, no obvious alternatives surfaced, no second opinion. There's no "Mint for AI tool spend."

You are not building a coding exercise. You are building something Techvruk could plausibly launch on similar Product Hunt next month with you making it us. Treat it that way.

What you're building

A free web app — call it whatever you want, naming is part of the test — that does this end-to-end:

1. A cold visitor lands on the page from a tweet, a blog post, or Hacker News
2. They input what AI tools they pay for, what plan, monthly spend, team size, and primary use case
3. They get an instant on-screen audit: where they're overspending, what to switch to or downgrade, and total potential monthly + annual savings
4. They get an option to capture the report (email gate) and — for high-savings cases — book a Techvruk expert consultation
5. The result is shareable via a unique public URL with proper Open Graph previews

No login required to use the tool. Email is captured *after* value is shown, never before.

MVP features (all six required)

1. Spend input form

Support at minimum these tools as of submission week:

Cursor (Hobby / Pro / Business / Enterprise)

- GitHub Copilot (Individual / Business / Enterprise)
- Claude (Free / Pro / Max / Team / Enterprise / API direct)
- ChatGPT (Plus / Team / Enterprise / API direct)
- Anthropic API direct
- OpenAI API direct
- Gemini (Pro / Ultra / API)
- Windsurf or v0 — your pick of one more

For each tool: which plan, current monthly spend, number of seats. Plus team size and primary use case (coding / writing / data / research / mixed). Form state must persist across page reloads.

2. Audit engine

For each tool the user is on, the engine evaluates:

- Are they on the right plan for their usage? (e.g., Team for 2 users is overkill)
- Is there a cheaper plan from the same vendor that fits?
- Is there a substantially cheaper alternative tool with similar capability for their use case?
- Are they paying retail when they could get the same thing through credits?

The logic must be **defensible**. A finance person should read your reasoning and agree. Not “Cursor bad, Claude Code good” — actual usage-fit reasoning with numbers.

Pricing data must be **current as of your submission week**. Sources cited in `PRICING_DATA.md` — every number must trace back to an official pricing page URL.

3. Audit results page

- Per-tool breakdown: current spend → recommended action → savings + 1-sentence reason
- Hero: total monthly savings + total annual savings, big and clear
- For audits showing >\$500/mo savings: surface Techvruk prominently as the way to capture more of that savings
- For audits showing <\$100/mo or already-optimal: be honest. “You’re spending well.” Don’t manufacture savings. Still capture the lead with a “notify me when new optimizations apply to your stack” signup

Visual quality matters. This is the page that gets screenshotted and shared.

4. AI-generated personalized summary

Use the Anthropic API (preferred — apply for free credits if you don’t have access) or any LLM to generate a ~100-word personalized summary paragraph based on the audit. Must handle API failures gracefully (fallback to a templated summary). Your full prompt goes in `PROMPTS.md`.

This is the *one* feature where you must use AI. For the audit math itself, hardcoded rules are correct — knowing when *not* to use AI is part of the test.

5. Lead capture + storage

- Email capture with optional fields: company name, role, team size
- Stored in a real backend: Supabase, Firebase, Cloudflare D1, your own Postgres on Render — your call
- Sends a transactional email (Resend / Postmark / SES free tier) confirming the audit and noting Techvrk will reach out for high-savings cases
- Basic abuse protection: rate limit, honeypot, or hCaptcha. Document your choice and why.

6. Shareable result URL

Each audit gets a unique public URL

- Identifying details (company name, email) stripped from the public version. Tools and savings numbers shown.
- Open Graph tags for clean link previews (Twitter card too)
- This is the viral loop — design accordingly

Bonus (only attempt after MVP works end-to-end)

PDF export of the full report

- Embeddable widget version (`<script>` tag a blogger could drop in)
- Benchmark mode: “your AI spend per developer is \$X — companies your size average \$Y”
- Referral codes — share the tool, both parties get a perk
- A short blog post or Twitter thread draft pitching the tool, written as if you were launching it.

Constraints

Frontend framework: React, Next.js, Vue, Svelte, SolidJS, or vanilla. Justify it in

ARCHITECTURE.md .

- **TypeScript strongly preferred.** If you use plain JavaScript, justify it.
- **No website builders.** No Wix, Webflow, Framer, Bubble. No “admin dashboard templates” where the UI is pre-built. Tailwind, shadcn/ui, MUI, Mantine, headless primitives are all fine.
- **Lighthouse mobile scores on the deployed URL:** Performance ≥ 85 , Accessibility ≥ 90 , Best Practices ≥ 90 .

No secrets in the repo. Use environment variables.

Deliverables

Submit a **single Google Form response** with the four items below. Anything missing = automatic rejection.

1. Public GitHub repo URL

Public. Contains everything in section 3.

2. Live deployed URL

Vercel, Netlify, Cloudflare Pages, Render, Fly.io, or equivalent. Localhost screenshots do not count. The URL must be reachable when we open it.

3. Required files at the repo root

LLMs read these files during evaluation. **Filenames and format are not optional.** All in plain markdown.

Engineering files

README.md

- 2–3 sentence summary of what you built and who it’s for
- 3+ screenshots or a 30-second screen recording (YouTube/Loom link)
- Quick start: install, run locally, deploy
- A **“Decisions”** section listing 5 trade-offs you made and why

- Link to the deployed URL

ARCHITECTURE.md

- A system diagram in **Mermaid** (renders inline on GitHub) or ASCII
- Data flow: how a user's input becomes an audit result
- Why you chose your stack
- What you'd change if this had to handle 10k audits/day

DEVLOG.md — the most important file we read

One entry per day, for 7 days. Backdating is obvious in git history; we check. Use this exact format:

```
## Day 1 — YYYY-MM-DD
**Hours worked:** X
**What I did:** ...
**What I learned:** ...
**Blockers / what I'm stuck on:** ...
**Plan for tomorrow:** ...
```

If you took a day off, write that entry too — Hours worked: 0 , with a one-line reason. Honesty scores higher than fake entries. We can tell.

REFLECTION.md

Answer all 5 questions, **150–400 words each** :

1. The hardest bug you hit this week, and how you debugged it (be specific — what hypotheses did you form, what did you try, what worked?)
2. A decision you reversed mid-week, and what made you reverse it
3. What you would build in week 2 if you had it
4. How you used AI tools (which tool, for what tasks, what you didn't trust them with, and one specific time the AI was wrong and you caught it)
5. Self-rating on a 1–10 scale for each: discipline, code quality, design sense, problemsolving, entrepreneurial thinking — with a one-sentence reason for each

TESTS.md

List every automated test you wrote: filename, what it covers, how to run it

- Minimum: 5 tests covering the audit engine specifically. They must actually run. We will run them.

[.github/workflows/ci.yml](#)

A GitHub Actions workflow that runs lint + tests on every push to main . Must show green checks on your latest commit.

PRICING_DATA.md

Your sources for every tool's pricing. Every number in your audit engine must trace to a URL on the vendor's official pricing page, with the date you pulled it. Format:

Cursor

- Pro: \$20/user/month – <https://cursor.sh/pricing> – verified YYYY-MM-DD
- Business: \$40/user/month – ...

PROMPTS.md

The full LLM prompts you used in the tool. Why you wrote them this way. What you tried that didn't work.

Entrepreneurial files

These are evaluated as carefully as the code. **Half of strong applicants under-invest here. Don't be them.**

GTM.md (300–700 words)

- Who is the *exact* target user — not “startups,” but a specific job title at a specific company stage
- What they Google or scroll through right before they'd want this tool
- Where they hang out online (specific subreddits, Slack groups, Discord servers, X lists)
- How you'd get the first 100 users in 30 days with \$0 paid budget — be specific, not “post on Twitter”
- The unfair distribution channel — what's the one thing only you could do
- What week-1 traction looks like if this works (specific numbers)

What scores well: specific, weird, real channels. What scores poorly: “we'll do SEO and content marketing.”

ECONOMICS.md (300–700 words)

If Techvruk deployed this tool tomorrow, run the unit economics:

- What's a converted lead worth to techvruk (estimate; show your reasoning)
- What's CAC at each channel from your GTM plan
- What conversion rate from “audit completed” → “Techvrk consultation booked” → “credit purchase” makes this profitable
- What would have to be true for this tool to drive \$1M ARR in 18 months
- Show the math even if your inputs are rough estimates. Approximate numbers > no numbers.

What scores well: a realistic spreadsheet-style breakdown. What scores poorly: vague TAM hand-waving.

USER_INTERVIEWS.md (3 interviews, ~150–300 words each)

Notes from **three real conversations** you had with potential users. 10–15 minutes each. Cold DM founders on X, ask in indie hacker Slacks, use your college network, talk to friends running side projects. For each:

- Name (or initials if they preferred anonymity), role, company stage
- 3+ direct quotes
- The most surprising thing they said
- What it changed about your design

These conversations must have happened. Fabricated interviews are obvious — they're generic, lack specific contradictions, and have no surprising moments. We've read enough of them to spot the pattern. Faking this is an instant reject. Talking to three humans this week is non-negotiable.

LANDING_COPY.md

Treat this as the actual landing page copy a marketer would ship:

- Hero headline (≤ 10 words)
- Subheadline (≤ 25 words)
- Primary CTA copy
- Social proof block (mocked is fine, indicate it's mocked)
- FAQ — 5 real Q&As

METRICS.md (200–500 words)

- Your single North Star metric — and why
- 3 input metrics that drive the North Star
- What you'd instrument first
- What number triggers a pivot decision

What scores well: metrics that match a B2B lead-gen tool at this stage. What scores poorly: “DAU” as a metric for a tool people use once a quarter.

4. Git history requirements

We read `git log`. These rules are checked programmatically:

- **Commits on at least 5 distinct calendar days** within the 7-day window.
Verify yourself with:

```
git log --pretty=format:@"%ad" --date=short | sort -u | wc -l
```

If fewer than 5, submission is rejected.

- **Conventional Commits format preferred** (feat: , fix: , docs: , refactor: , chore: , test:)
- **Meaningful commit messages.** “update”, “fix”, “wip”, “asdf” are red flags. “fix: handle 429 from Anthropic API gracefully in summary fallback” is what we want.

Evaluation rubric (100 points total)

Dimension	Weight	What scores well
Entrepreneurial thinking	25	GTM, economics, user interviews, landing copy, metrics — all show real founder mindset, not template-fill
Engineering skills	15	Git hygiene, CI green, ≥ 5 working tests, deploy works, accessibility considered
Thinking models	15	ARCHITECTURE depth, REFLECTION specificity, README “Decisions” section non-trivial
Programming skills	15	Code is readable, sensible abstractions, types used well, no obvious bugs in happy path
Hard work	10	All 6 MVP features work, polish in UI, bonus attempted
Discipline & consistency	10	DEVLOG has 7 dated entries with depth, commits across ≥ 5 days
Polish of the audit logic itself	10	A finance-literate person reads your reasoning and agrees with it

Ground rules

- **AI tools (Cursor, Claude, ChatGPT, Copilot, Codex, etc.) are allowed and expected.** Techvruk is an AI infrastructure company. Disclose your usage in `REFLECTION.md` honestly. We can tell when an entire codebase is one-shot generated, and that autorejects — not because AI is bad, but because a one-shot codebase tells us nothing about *you*.
- **The user interviews requirement is real.** Three actual humans. The quality of these conversations is one of the strongest signals we read. They are also the easiest part to explain to customers when we hire you.
- **Pricing data accuracy matters.** Every number must cite a vendor URL. We spot-check.
- **No private dependencies, no closed-source libraries, no hardcoded secrets.**
- **No communication during the week** beyond the email below. Part of what we evaluate is how you handle ambiguity. Make a reasonable assumption, document it in `DEVLOG.md`, move on.
- **By submitting, you grant Techvruk a non-exclusive license to learn from public elements of your submission.** You retain full ownership of your code. Use it for your portfolio.

What happens next

- Within **3 working days** of the deadline, you'll hear back: shortlisted (Round 2), or not.
- **Round 2** is a smaller, focused build (~2 days, more constrained scope). Released to shortlisted candidates only.
- **Interviews** follow Round 2 — 1–2 conversations with the team, including a walkthrough of your Round 1 + Round 2 code, and a discussion of one decision in your `ECONOMICS.md` or `GTM.md`.

We're a small team building a real product in a noisy market. We hire people who ship, write down their thinking, and treat their own work seriously. Show us you're one of them.

Good luck.

— Techvruk